

# **Implementation of making of a Selective Compliance Assembly Robot Arm (SCARA)**

MINI PROJECT REPORT

Submitted by

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## **Certificate**

This is to certify that the Mini project entitled **Implementation of making of a Selective Compliance Assembly Robot Arm (SCARA)** is a bonafide work carried out by **AMISHA SARUP A (KTE23RAI008)**, **ASWATHI ANIL (KTE23RAI011)**, **AYISHA ZIYA (KTE23RAI014)**, **FATHIMATH SHANA AP (KTE23RAI019)** during 2026-27, in partial fulfilment for the award of the B. Tech degree in Robotics and Artificial Intelligence of APJ Abdul Kalam Technological University, Rajiv Gandhi Institute of Technology, Kerala.

**Guide**

**Co-ordinators**

**Head of the Department, ECE**

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## **ABSTRACT**

This project focuses on the design and implementation of a 4-DoF SCARA (Selective Compliance Articulated Robot Arm) robot for automated drawing of arc based shapes and pattern applications. SCARA robots are widely recognized for their high speed, precision, and excellent planar motion capabilities, making them ideal for tasks requiring accuracy and repeatability. In this project, the robot is specifically developed to perform drawing arcs and patterns on a paper surface, such as labeling, and shape drawing. The robot is designed with optimized structural and mechanical parameters to ensure smooth horizontal motion and stable vertical control for consistent pen pressure. A combination of stepper and servo motors is used to achieve precise motion control across all joints. The system is powered through an efficient power supply setup and controlled using an Arduino-based architecture integrated with motor drivers, enabling synchronized and accurate operation.

A key advantage of this system is its ability to automate repetitive drawing arc based tasks, thereby reducing human effort, minimizing errors, and improving consistency. The programmable nature of the robot allows continuous operation with high reliability, making it suitable for educational, laboratory, and light industrial automation applications. Overall, the project demonstrates an effective integration of mechanical design, electronics, and control systems to achieve a cost-efficient and high-performance robotic solution.

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# CHAPTER-1

## INTRODUCTION

Automation has become an essential part of modern industries and research applications, enabling improved efficiency, precision, and repeatability. In fields such as manufacturing, graphics, and educational robotics, accurate drawing and positioning tasks play a significant role. Traditional manual methods for drawing and pattern generation are time-consuming, prone to human error, and lack consistency. To address these limitations, robotic solutions such as SCARA (Selective Compliance Assembly Robot Arm) robots can be used due to their high speed, precision, and reliability in planar motion.



SCARA robots are widely recognized for their high speed, rigidity, and accuracy in planar motion, making them highly suitable for applications that require precise positioning. Their ability to perform controlled movements in the horizontal plane, combined with simple vertical actuation, makes them ideal for tasks such as drawing arcs and pattern generation.

This project focuses on the design and development of a 4-DOF SCARA robot optimized for automated drawing arc based applications. The robot is designed to achieve smooth and accurate trajectory tracking in the X-Y plane, along with controlled Z-axis motion for pen lifting and placement. By integrating an efficient mechanical structure with microcontroller-based control systems, the proposed SCARA robot aims to improve precision, repeatability, and ease of operation. The system provides a cost-effective and practical solution for demonstrating robotic automation in graphical and educational applications.

### 1.A REAL TIME PROBLEM

Manual drawing arcs and pattern generation tasks often face many challenges like lack precision, repeatability, and speed, especially in applications like graphical plotting and repetitive pattern generation. including:

- Time-Consuming Processes: Manual drawing of shapes, patterns, or text requires significant time and effort, especially for repetitive tasks.
- Inconsistent Precision: Human drawing lacks uniform accuracy and repeatability, leading to variations in output quality.

- Limited Repeatability: Reproducing the exact same pattern multiple times is difficult without automation.
- Human Fatigue: Continuous drawing tasks can lead to fatigue, reducing efficiency and increasing errors.
- Lack of Automation: Traditional methods do not allow programmable control or integration with digital systems.

To address these limitations, a SCARA-based drawing robot is developed to perform accurate and repeatable operations using controlled X–Y motion and a Z-axis pen mechanism, reducing manual effort and improving efficiency..

## **1.B OBJECTIVE**

The objective of the SCARA system is to develop a user-friendly and cost-effective robotic arm capable of performing precise and repeatable drawing arc based patterns, suitable for educational, graphical, and automation-based applications.

## **1.C SOLUTION**

To address precision and repeatability challenges in drawing arc applications, this project presents a 4-DOF SCARA robot with an optimized mechanical and control system. The robot features a lightweight yet rigid structure designed for smooth planar motion, along with a lead screw–based Z-axis mechanism for precise pen actuation. Servo motors and a stepper motor enable accurate and coordinated movement, while an Arduino-based control system ensures reliable operation. The system follows predefined trajectories to perform drawing arc based tasks efficiently, making it suitable for educational, graphical, and automation-based applications.

## **1.D SCOPE OF PROJECT**

This project covers multiple aspects of SCARA robot design, development, and implementation for drawing arc based patterns, including:

- Mechanical Design: Developing a lightweight and rigid robotic arm optimized for smooth and accurate planar motion.
- Kinematic Analysis: Studying the motion of the SCARA robot to achieve precise positioning in the X–Y plane.
- Hardware Selection & Integration: Integrating motors, drivers, and controllers for reliable and coordinated movement.
- Software Development: Implementing control algorithms for trajectory planning and precise drawing operations.
- Testing & Performance Evaluation: Evaluating accuracy, repeatability, and smoothness of drawing.
- Applications: Demonstrating the use of the SCARA robot in drawing arcs, pattern generation, and educational visualization.

By implementing this 4-DOF SCARA arc based drawing robot, the system enhances precision, reduces manual effort, and provides a cost-effective solution for automated drawing and educational robotics applications.



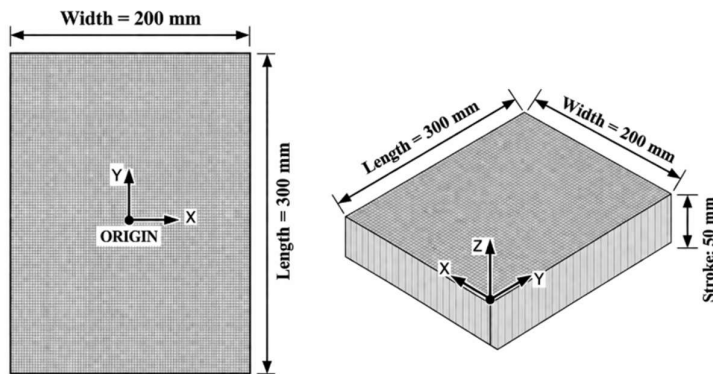
## CHAPTER-2

### DESIGN PARAMETERS

#### 2.A WORKSPACE

The workspace of a SCARA (Selective Compliance Articulated Robot Arm) defines the area within which the robot can perform drawing operations effectively. It is a crucial factor in determining the robot's ability to cover the required drawing surface, such as an A4 sheet. The workspace is influenced by link lengths, joint limits, degrees of freedom (DOF), and the reach of the end-effector.

The SCARA robot designed for drawing applications has a horizontal reach of approximately 0 to 300 mm, enabling it to cover a typical drawing area efficiently. The vertical stroke is limited to around 50 mm, which is sufficient for pen lifting and placement. This workspace configuration provides an optimized and controlled area suitable for precise, smooth, and repeatable drawing and writing tasks.



Maximum horizontal reach: 0 to 300 mm  
Vertical stroke: 50 mm  
Workspace: A4-size (300 mm × 200 mm × 50 mm) (X × Y × Z)

#### 2.B Task Planned

The primary task of this SCARA robot is automated drawing arcs and patterns, where it accurately traces predefined shapes and patterns with high precision and repeatability. Its rigid structure and flexible planar motion enable smooth and efficient operation.

Key Aspects of the Drawing Task:

- High Precision – Accurate reproduction of shapes and patterns
- Repeatability – Consistent output for multiple drawings.
- Smooth Motion – Controlled trajectory in the X–Y plane
- Automation Efficiency – Reduces manual effort and improves consistency

This system enhances drawing accuracy and provides a reliable solution for automated graphical applications.

## 2.C Structural parameters

The structural design of the SCARA robotic arm plays a crucial role in determining its efficiency, stability, and operational capability. The following key structural parameters were considered during the design and development phase:

### 2.C.i. Materials Used

The materials used in the construction of the SCARA robotic arm have been carefully selected to ensure durability, efficiency, and sustainability. Forex sheets are utilized for the body, providing a lightweight yet sturdy structure. steel rods are purposed as the supporting framework, enhancing strength. Additionally, PVC components are used to support the links, ensuring smooth and frictionless movement between the connected sections, thereby improving the overall efficiency of the robotic arm.



### 2.C.ii Type of joints

The robot consists of a combination of revolute and prismatic joints, specifically:

- Two revolute joints (R-joints): These joints allow rotation around fixed axes, enabling movement of the arm in the X–Y plane within a circular workspace.
- One prismatic joint (P-joint): This joint provides linear motion in the vertical direction, allowing controlled pen up and down movement during drawing.
- One additional revolute joint (R-joint): This joint enables orientation of the pen for improved control and flexibility during drawing tasks.

This RRPR configuration provides the robot with four degrees of freedom (4-DOF), ensuring precise positioning, smooth trajectory execution, and enhanced control for drawing operations.

### 2.C.iii Link length

The link lengths were selected and optimized based on workspace requirements and structural stability:

- First Link Length (L1): 140 mm
- Second Link Length (L2): 110 mm
- Third Link Length (L3): 130 mm
- Forth Link Length (L4): 50 mm
- Total Horizontal Reach: 0 mm to 250 mm
- Vertical Stroke (Prismatic Joint): 50 mm

These link dimensions ensure that the SCARA robot provides adequate coverage for typical drawing surfaces such as an A4 sheet. The design offers a compact and stable workspace suitable for precise, smooth, and repeatable drawing arc based pattern operations.

## 2.D Degree of Freedom

The proposed SCARA (Selective Compliance Articulated Robot Arm) robot has four degrees of freedom (4-DOF):

1. Base Rotation : Controls the rotation of the entire arm, enabling movement across the workspace.
2. Elbow Rotation : Provides additional motion in the X–Y plane for precise positioning.
3. Linear Motion : Enables vertical movement of the end-effector for controlled pen up and down action.
4. Wrist Rotation: Allows orientation of the pen for improved flexibility during drawing tasks.

This configuration enables precise and smooth motion in the X–Y plane with controlled Z-axis movement, making the SCARA robot well-suited for accurate drawing and writing applications..

## 2.E DH parameters

DH Parameter Table (Standard Convention)

| Link (i) | $\theta_i$ | $d_i$ | $a_i$  | $\alpha_i$ | Joint Type |
|----------|------------|-------|--------|------------|------------|
| 1        | $\Theta_1$ | 0     | 140 mm | 0          | Revolute   |
| 2        | $\Theta_2$ | 0     | 110mm  | 0          | Revolute   |
| 3        | $\Theta_3$ | $d_3$ | 0      | 0          | Prismatic  |
| 4        | $\Theta_4$ | 50    | 0      | 0          | Revolute   |

$\theta_i$ : Variable for revolute joints, fixed (0) for prismatic

$d_i$ : Variable for prismatic joint (Joint 2), fixed (0) for revolute

$a_i$ : Link lengths along X-axis between joints

$\alpha_i$ :  $0^\circ$  since all joints rotate/translate in the same horizontal plane (no twist between joint axes)

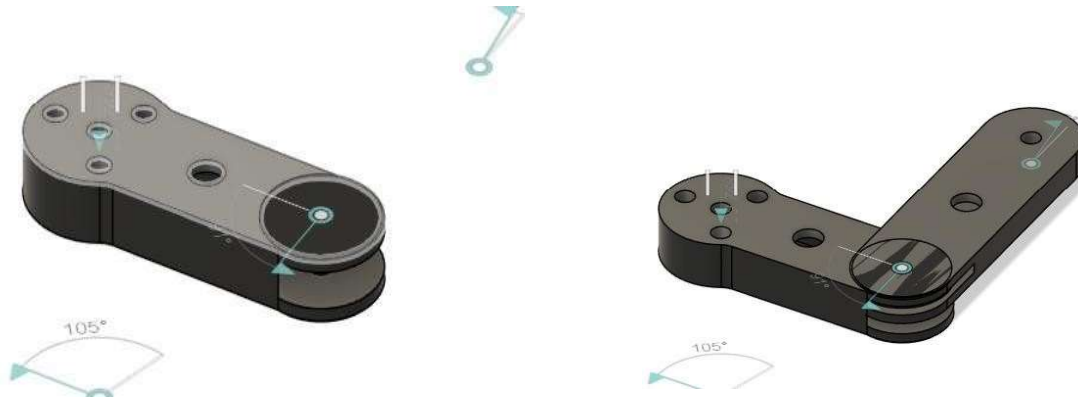
## 2.F Drawing



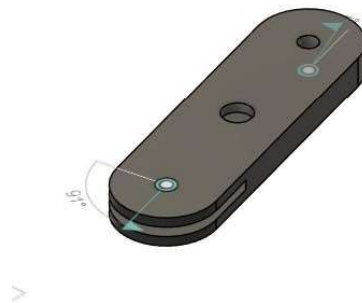
Link 1



Link 2



Link 3



Link 4



## Chapter 3

### MECHANICS/ MECHANICAL PARAMETERS

#### 3.A Materials Used

##### 1. LEAD SCREW



In our SCARA robot, a **lead screw mechanism** is used to achieve vertical (Z-axis) motion required for pen control. The lead screw converts the rotational motion of the motor into precise linear movement, allowing the pen to move up and down accurately during drawing arc based pattern operations. This mechanism ensures smooth, controlled motion for proper pen contact with the surface, improving drawing quality and consistency. It also provides good stability and rigidity, which is essential for maintaining precision in drawing arcs and arc based patterns.

##### 2. COUPLER

In our SCARA robot, a **6.35 mm to 8 mm flexible shaft coupler** is used to connect the NEMA 23 motor shaft to the lead screw. This coupler enables efficient transmission of rotational motion from the motor to the lead screw, producing smooth and precise vertical (Z-axis) movement. The flexible design helps accommodate minor misalignment between the motor and lead screw, reducing vibrations and preventing mechanical stress. This ensures reliable performance and improves the overall accuracy of the vertical motion system, which is essential for consistent arc drawing

operations.



### 3. BEARING

The SCARA robot utilizes both linear and radial bearings to ensure smooth and stable mechanical operation:

- **Linear Bearing:** Positioned between the steel rod and Link 2, the linear bearing allows for frictionless vertical motion along the Z-axis. It guides the movement of Link 2 with high precision, ensuring stability and smooth sliding action during the lead screw-driven vertical displacement.
- **Radial Bearings:** These are mounted at the rotational joints of the SCARA arm to support shafts undergoing circular motion. They reduce friction during rotation and help maintain alignment, allowing for smooth and controlled angular movement of the links.

The combined use of these bearings improves mechanical efficiency, reduces wear, and contributes to the overall reliability of the SCARA mechanism.



### 3.B Kinematics

The Selective Compliance Articulated Robot Arm (SCARA) features a configuration well-suited for precise planar motion, making it ideal for applications such as arc drawing. Its kinematic structure consists of two rotational joints (shoulder and elbow) for movement in the X–Y plane, a prismatic or linear joint for vertical (Z-axis) motion, and an optional rotational joint at the end-effector for orientation. In terms of kinematics:

- Forward kinematics involves calculating the end-effector position based on given joint angles and link lengths. This is achieved using standard transformation matrices applied sequentially through each joint and link.

$$x = L_1 \times \sin(\theta_1) + L_2 \times \sin(\theta_1 + \theta_2)$$

$$y = L_1 \times \cos(\theta_1) + L_2 \times \cos(\theta_1 + \theta_2)$$

- Inverse kinematics is the process of determining the required joint angles to position the end-effector at a desired location. For SCARA robots, this can often be solved analytically due to their relatively simple planar structure.

$$\theta_2 = \arccos((x^2 + y^2 - L_1^2 - L_2^2) / 2 \times L_1 \times L_2)$$

$$\theta_1 = \arctan(x / y) - \arctan((L_2 \times \sin(\theta_2)) / (L_1 + L_2 \times \cos(\theta_2)))$$

The SCARA robot offers high rigidity in the vertical direction and compliance in the horizontal plane, allowing smooth and accurate motion. This makes it highly suitable for applications requiring precision, repeatability, and controlled trajectory execution, such as automated arc drawing and pattern generations.

### 3.C. Dynamic Parameters

The dynamic analysis of a SCARA robot deals with understanding how forces and torques affect the motion of its links and joints. These parameters are crucial for designing the actuators, selecting appropriate materials, and ensuring safe and efficient motion control.

Key dynamic parameters include:



- **Mass and Inertia of Links:** Each link has a mass and moment of inertia that influence how much torque is required to accelerate or decelerate during movement. Lighter links reduce energy consumption and increase speed.
- **Joint Torques:** Dynamic modeling calculates the torque required at each joint to achieve a desired motion trajectory, taking into account the effects of gravity, Coriolis, and centrifugal forces.
- **Friction and Damping:** Joint friction and internal damping impact the smoothness of motion and energy efficiency. Proper compensation is necessary for precise control.
- **End-effector load:** The payload is limited to a lightweight pen, resulting in minimal impact on the system's dynamics and allowing for stable and precise operation.

### **3.D. Trajectory Planning**

Trajectory planning involves generating a smooth and continuous path for the SCARA robot's end-effector to follow from a starting point to a target position, while maintaining constraints on position, velocity, and acceleration.

In SCARA drawing robots, trajectory planning is typically performed in Cartesian space to accurately trace shapes and patterns. Common techniques include:

- **Linear Interpolation:** Used to draw straight lines between two points with controlled speed.
- **Cubic or Quintic Polynomial Planning:** Provides smooth motion by controlling velocity and acceleration, ensuring high-quality drawing without jerks.
- **Trapezoidal Velocity Profiles:** Enables smooth acceleration and deceleration, improving motion stability and reducing mechanical stress.

Effective trajectory planning is essential to maintain mechanical stability, reduce wear and tear, and achieve high-speed, precise operations without violating dynamic limits such as joint torque or velocity constraints.

## Chapter 4

### ELECTRICAL/ ELECTRONIC CONTROL

#### 4. A Design of Motors

The motor selection for the SCARA drawing robot is based on achieving precise, smooth, and coordinated motion in the X–Y plane along with controlled Z-axis movement. Unlike conventional designs that use only servo motors, this system uses a combination of a stepper motor and servo motors for optimal performance.

The motors used in the system are:

- **NEMA 23 Stepper Motor (Base Joint):** Provides precise rotational control for the base and up and down motion, enabling accurate positioning in the X–Y plane.
- **MG996R Servo Motor (Elbow Joint):** Controls the second rotational joint for smooth planar motion.
- **MG90 Servo Motor (Wrist):** Controls pen orientation if required.

This combination ensures accurate trajectory tracking, smooth motion, and high repeatability for drawing operations

#### Assumptions

- End effector weight: 50g (0.050 Kg)
- Link weights (approx):
  1. Link 1: 0.20 Kg
  2. Link 2: 0.15 Kg
  3. Link 3: 0.28 Kg
  4. Link 4: 0.10 Kg

Torque calculation and motor selection

- End effector : 50g
- Servo motor1(end effector motor : SG90) weight :20g

1. Force from the weights:

$$F = \text{mass} \times \text{gravity} (9.81 \text{ m/s}^2)$$

- Object Force (50g):  
 $F = 0.05 \text{ kg} \times 9.81 = 0.491 \text{ N}$
- Link 1 (0.2 kg):  
 $F1 = 0.2 \text{ kg} \times 9.81 = 1.962 \text{ N}$
- Link 2 (0.15 kg):  
 $F2 = 0.15 \text{ kg} \times 9.81 = 1.472 \text{ N}$
- Link 3 (0.28 kg):  
 $F3 = 0.2 \text{ kg} \times 9.81 = 2.75 \text{ N}$
- Link 4 (0.10 kg):  
 $F4 = 0.1 \text{ kg} \times 9.81 = 0.981 \text{ N}$

## 2. Torque from each part:

$$\text{Torque } \tau = \text{Force} \times \text{Distance (from joint)}$$

- Joint 1 :  
 $\tau_3 = (0.20 \times 9.81 \times 0.07) + (0.15 \times 9.81 \times 0.195) + (0.28 \times 9.81 \times 0.25) + (0.10 \times 9.81 \times 0.275)$   
 $= 1.3808 \text{ Nm}$   
 Applying safety margin as 1.5x ; Torque 2

**NEMA 23(30-60 RPM)** is used.

- Joint 2:  
 $\tau_2 = (0.15 \times 9.81 \times 0.055) + (0.28 \times 9.81 \times 0.11) + (0.10 \times 9.81 \times 0.135)$   
 $= 0.5156 \text{ Nm}$   
 Applying safety margin as 1.5x ; Torque 2

**MG996R(40-50 RPM)** is used

- Joint 3:  
 $\tau_3 < 0.05 \text{ N}$   
 Applying safety margin as 1.5x ; Torque 2  
**MG90 micro servo (80-100 RPM)** is used

## 3. Speed of each Motor

Motor1: NEMA23 : 60 RPM

Motor2: MG996R : 50 RPM

Motor3: MG90S : 100 RPM

## B.Motors used

### 1. NEMA 23 Stepper Motor



Provides high torque and precise position control for the base rotation, up and down motion. Suitable for accurate angular movement in the X–Y plane.

### 2. MG996R Servo Motor



The MG996R servo motor is a high-torque, metal gear servo widely used in robotic applications requiring strong and reliable motion. It operates at 4.8–7.2V and provides high torque (up to ~11 kg·cm).

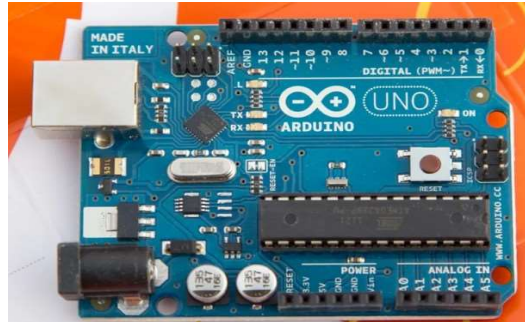
### 3. MG90 Micro Servo



The MG90 micro servo motor is a compact and lightweight actuator used for precise angular positioning in robotic applications. It operates typically at 4.8–6V.

## 4.C Details of control system

### Arduino UNO



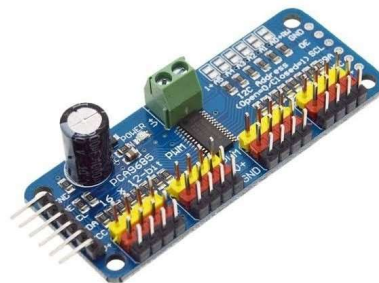
The Arduino UNO is an open-source microcontroller board based on the ATmega328P. It is used to process inputs and control outputs of motors. It offers digital and analog I/O pins, making it ideal for robotic control systems.

### TB6600 Stepper Motor Driver



The motor driver acts as an interface between the microcontroller and the NEMA23 motor. It supplies sufficient current and voltage to drive stepper. It enables control over direction and speed through input signals from the Arduino UNO.

### PCA9685 Servo Driver Module



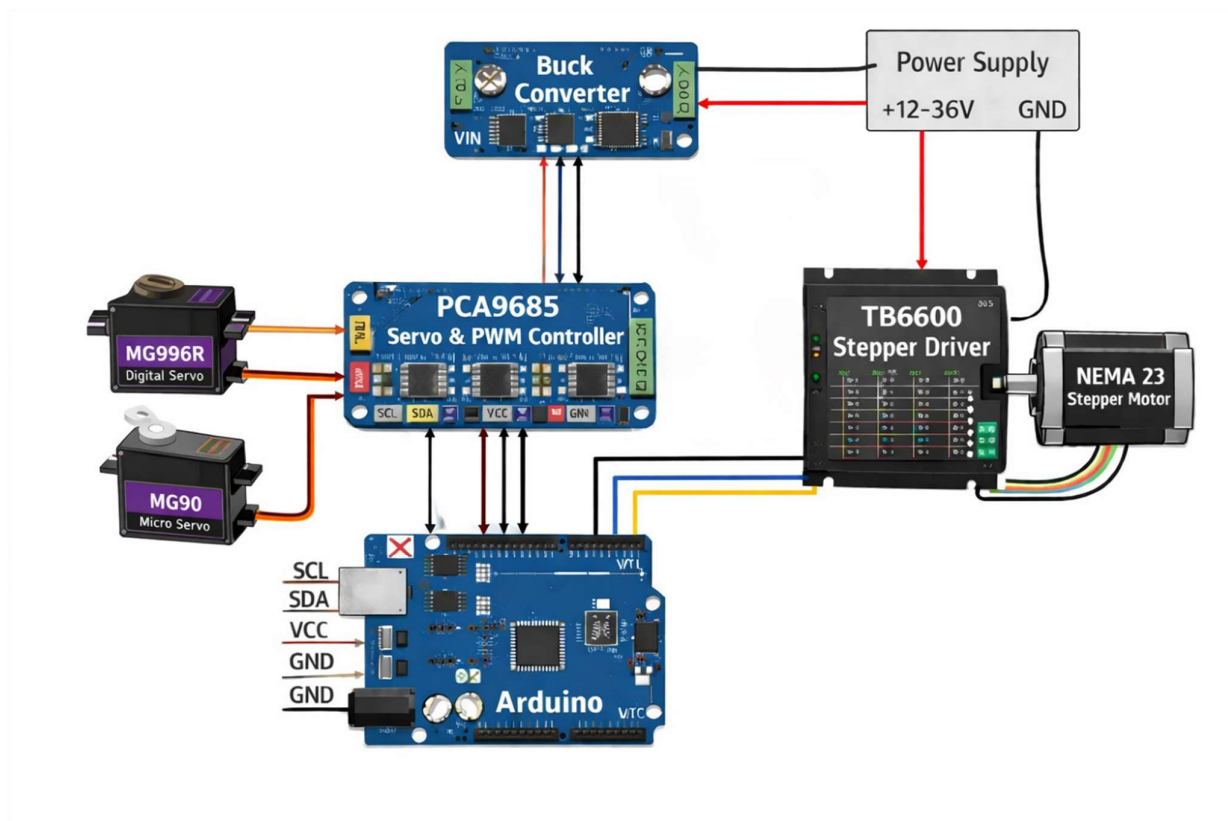
The motor driver acts as an interface between the microcontroller and the motors. It supplies sufficient current and voltage to drive servo motors. It enables control over direction and speed through input signals from the Arduino UNO.

#### DC-DC buck converter (24V → 6V)



A DC-DC buck converter is used to step down a higher DC voltage to a lower, stable output voltage efficiently. In this SCARA robot, it converts 24V from the power supply to a regulated 5–6V suitable for powering servo motors and control electronics.

#### 4.C Circuit Diagram



## **Chapter 5**

### **SOFTWARE IMPLEMENTATION**

#### **5.A Trajectory-Based Control System**

##### Overview

The SCARA robot is controlled using predefined trajectories for drawing arc based shapes and patterns. Instead of manual interfaces, the system follows programmed paths to achieve accurate and repeatable motion.

##### Functionality

- Path Execution: The robot follows predefined coordinates to draw arc based shapes and patterns.
- Coordinated Motion: Base and elbow joints move simultaneously for smooth X–Y motion.
- Z-Axis Control: A servo motor controls pen up/down movement.
- Motion Limits: Joint limits are defined to ensure safe operation.

##### Implementation

The control system is implemented using an Arduino UNO. The NEMA 23 stepper motor is driven using a TB6600 driver, while servo motors are controlled via a PCA9685 module. Basic inverse kinematics is used to convert drawing coordinates into joint angles.

##### Usability and Performance

The system ensures smooth and accurate drawing with minimal user intervention. Predefined paths allow consistent and repeatable outputs.

#### **5.B Kinematic Control**

##### Overview

Kinematic control is used to determine the position of the end-effector based on joint angles. This enables accurate movement of the pen in the X–Y plane.

### Functionality

- Forward Kinematics: Calculates end-effector position from joint angles.
- Inverse Kinematics: Computes joint angles required for a desired position.
- Smooth Motion: Ensures continuous path following without jerks.

### Implementation

Mathematical equations are used to compute  $\theta_1$  and  $\theta_2$  for given X–Y coordinates. These values are then sent to the motors for execution.

### Usability and Performance

Kinematic control improves drawing accuracy and ensures consistent reproduction of patterns.

## **5.C Motor Control and Execution**

### Overview

Motor control is responsible for executing the calculated joint movements using stepper and servo motors.

### Functionality

- Stepper Control: Controls base rotation using TB6600 driver.
- Servo Control: Controls elbow, Z-axis, and wrist using PCA9685.
- Synchronization: Ensures all motors move in coordination.
- Speed Control: Maintains smooth motion during drawing.

### Implementation

Arduino generates control signals for both stepper and servo motors. PWM signals are used for servos, while pulse signals drive the stepper motor.

### Usability and Performance

The system provides stable and reliable motor control, enabling smooth trajectory execution and accurate drawing performance.



## Chapter 6

### DETAILS OF COST INVOLVED

| Components                       | Price       |
|----------------------------------|-------------|
| Forex Sheet                      | 305         |
| NEMA 23 Stepper Motor            | 1625        |
| MG90 micro Servo Motor           | 125         |
| 6.35*8mm flexible shaft coupling | 55          |
| DC–DC buck converter             | 195         |
| Lead screw and nut               | 762         |
| MG996R motor                     | 300         |
| Arduino UNO                      | 254         |
| Motor Driver (PCA9685)           | 229         |
| TB6600 stepper motor driver      | 395         |
| Miscellaneous                    | 519         |
| <b>Total Cost</b>                | <b>4764</b> |

## **Chapter 7**

### **CONCLUSION**

The development of the SCARA robot for automated drawing arc based shapes and patterns has been successfully implemented, providing an efficient and precise solution for trajectory-based operations. The system uses an Arduino-based control unit to coordinate the motion of stepper and servo motors, enabling smooth and accurate movement in the X–Y plane along with controlled Z-axis pen actuation. This allows the robot to reproduce shapes, and patterns with high precision and repeatability.

The system has demonstrated reliable performance with stable motor control and consistent drawing output. While minor limitations such as mechanical vibrations and alignment accuracy may affect performance, the overall system provides a significant improvement over manual drawing methods.

Future enhancements can include improved trajectory planning, better mechanical stability, and integration of advanced algorithms for more complex pattern generation. These improvements will further enhance the accuracy, efficiency, and versatility of the SCARA drawing robot for educational and automation applications.

## **Chapter 8**

### **RESULT**

#### **8.1 System Performance**

The SCARA robot successfully performs automated drawing arc based shapes and pattern operations using coordinated control of stepper and servo motors. The implementation of trajectory-based motion ensures smooth and accurate movement in the X–Y plane along with precise Z-axis pen control.

Key Achievements:

1. **Accurate Trajectory Execution:** The robot effectively follows predefined paths to draw shapes and patterns.
2. **Smooth Motion Control:** Coordinated joint movement ensures continuous and stable drawing.
3. **Reliable Motor Control:** Stepper and servo motors operate consistently with minimal error.
4. **Stable System Operation:** The control system ensures reliable communication and synchronized motion.

#### **8.2 Accuracy and Precision**

1. **Drawing Accuracy:** Achieves high positional accuracy in reproducing shapes and patterns.
2. **Repeatability:** The robot consistently reproduces the same drawing with minimal variation
3. **Motion Response:** The system responds quickly to control inputs, ensuring smooth trajectory execution.
4. **Z-Axis Control:** Precise pen up/down motion ensures clean start and stop of drawing.

## Chapter 9

### REFERENCES

- Urrea, C., & Pascal, J. (2016). Design, simulation, comparison and evaluation of parameter identification methods for an industrial robot. *Computers & Electrical Engineering*.
- Subhashini, P. V. S. (2017). Placement analysis of rectangular components within the work volume using a SCARA robot. *IEEE International Conference on Nascent Technologies in Engineering (ICNTE)*.
- Misra, A., & Singh, G. (2019). Kinematic and dynamic analysis of an industrial robotic manipulator. *International Conference on Robotics, Automation and Non-Destructive Evaluation*.
- Nof, S. Y. (1999). *Handbook of Industrial Robotics*. John Wiley & Sons.
- Liyanage, M., Rideout, G., & Krouglicof, N. (2012). Modeling and analysis of the dynamic performance of a high-speed SCARA robot. *International Conference on Bond Graph Modelling*.
- Kumra, S., Saxena, R., & Mehta, S. (2012). Design and development of robotic arm controlled by man-machine interface. *IEEE Conference Publications*.
- Cetinkunt, S. (2015). *Mechatronics with Experiments*. Wiley.
- Siciliano, B., & Khatib, O. (2008). *Handbook of Robotics (2nd ed.)*. Springer..
- Fang, J., Liu, J., & Wang, D. (2008). Design of SCARA robot motion controller. *Journal of Jilin Teachers Institute of Engineering and Technology* .